

Process FMEA — Part 45621-B (Turbine Bracket)

Rev A · effective 2024-11-04 · aerospace part (AS9100D applies). Customer: Aeronova (CUST-ANV).
Material: Inconel 718. Cross-reference: process spec `PLM-PS-INC718-0042`.

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--- METADATA ---
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title: Process FMEA — Part 45621-B — Rev A
domain: quality
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owner: L. Pereira, Quality Engineering — Wichita
part_numbers: [45621-B]
operation_numbers: [OP20, OP30]
machine_ids: [DMU-14, DMU-22]
controller_models: [Fanuc 0i-TF]
fault_codes: []
materials: [MAT-INC718]
standards: [AS9100D, IATF-16949-2016]
supplier_ids: []
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1. Scope

Process FMEA for OP20 (face & rough mill) and OP30 (finish 5-axis mill) of Part `45621-B` as run on `DMU-14` / `DMU-22`. Stock prep (OP10 on `BSAW-03`) is out of scope.

2. Failure mode table

ID	Operation	Failure mode	Effect	S	Cause(s)	O	Current control	D
FM-101	OP20 — Rough mill	Datum A non-flat (> 0.05 mm)	OP30 fixtures incorrectly; profile out of tolerance.	6	Insert wear; fixture chip.	3	CMM check first piece + every 10.	3
FM-104	OP30 — Finish 5-axis mill	Tool wear past limit causing profile out-of-	Scrap; potential warranty escape if	8	(a) Cumulative flank wear VB > 0.20 mm (the limit set for Inconel	5	Time-based tool change per <code>PLM-PS-INC718-0042</code> ; visual check at each tool	3

		tolerance ($> \pm 0.025$ mm)	not detected.		718 — see PLM-PS-INC718-0042). (b) Operator missed tool-change at 60-minute interval. (c) Through-spindle coolant pressure dropped below 18 bar.		change; CMM 100% at OP40. Note: tool wear does NOT raise Fanuc alarm 300; alarm 300 is a position-feedback fault (see PDM-MAN-FANUC-01TF-001 §7.3). Tool-wear shutdowns surface as program messages or operator-observed surface-finish degradation.	
FM-105	OP30	Pocket geometry undersize	Rework or scrap.	5	Tool diameter compensation entry error.	4	First-piece CMM; tool-table verification step in setup sheet.	3
FM-106	OP30	Burr at pocket edge	Deburr rework required; risk of FOD in assembly.	4	Worn finishing pass insert.	5	Visual at each piece.	4

3. Tool-wear linkage to process spec

The tool-wear failure mode (FM-104) is the primary risk on this part. The flank-wear limit $VB_{max} = 0.20$ mm and the 60-minute cumulative-cut tool-change rule are defined in PLM-PS-INC718-0042 . The PFMEA references those values rather than duplicating them.

Hint for cross-source queries: A query of the form “machine DMU-14 threw alarm 300 and the operator suspects tool wear” resolves across three sources: PDM-MAN-FANUC-01TF-001 (alarm 300 is position-feedback, not tool wear) + PLM-PS-INC718-0042 (the actual tool-wear limit for Inconel 718 on DMU-14) + this PFMEA Rev A (FM-104, tool-wear failure mode and current controls).

Document control / sign-off

Role	Name	Signature	Date
Document Owner	L. Pereira, Quality Engineering — Wichita	(on file)	2024-11-04
Approver	L. Pereira, Quality Engineering — Wichita	(on file)	2024-11-04